AI-Driven Intrusion Detection and Attention-Guided Root Cause Diagnosis for Cyber-Resilience in Power Grid Operational Technology Networks

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***Abstract*—Operational Technology (OT) networks in power grid substations are increasingly exposed to protocol compliant, coordinated cyberattacks that evade conventional detection while demanding rapid, targeted response. Most existing intrusion detection systems (IDS) for OT, provides only a binary anomaly decision, leaving operators without insight into which device or protocol is affected or what corrective action to take. This paper proposes an attention-guided, resilience-aware intrusion detection and diagnosis framework for multi protocol OT networks. The proposed framework formulates detection as a spatio temporal graph learning problem, in which an edge conditioned graph neural network (GNN) models cross protocol device interactions and a recurrent attention mechanism captures their temporal evolution to flag deviations from learned normal behavior. Building on this detection layer, we propose an attention-guided root cause and localization mechanism that attributes each anomaly to a specific device, protocol, and time window, followed by a resilience action mapping layer that translates the localized signature into a MITRE ATT&CK aligned, actionable mitigation by operator. A detailed hardware in the loop (HIL) substation testbed has been developed, and a labeled multi protocol dataset covering seven deterministic and dynamic attack classes has been generated using this testbed. The proposed methodology of detection, diagnosis, and response presented here is the ongoing effort, moving OT intrusion detection from a binary alert toward an interpretable, response ready decision to enable cyber-resilience and quantifying through metric.**

***Index Terms*—Operational Technology, Intrusion Detection, Explainable AI, Root Cause Diagnosis, Incident Response, Graph Neural Networks, Cyber Physical Resilience.**

I. Introduction

The increasing integration of communication protocols such as DNP3 and IEC 61850 (GOOSE, Sampled Values) into substation automation has expanded the attack surface of power system OT networks, as underscored by recent incidents targeting energy infrastructure [1]. These environments differ fundamentally from IT networks in strict timing constraints, protocol specific semantics, and tightly coupled cyber physical interactions, which make both attack execution and attack detection highly context dependent. Rule based IDS approaches offer strong interpretability by enforcing protocol invariants but are limited to deterministic violations, while data driven models capture complex, coordinated behavior but typically operate as opaque, binary classifiers [2]. In earlier work,

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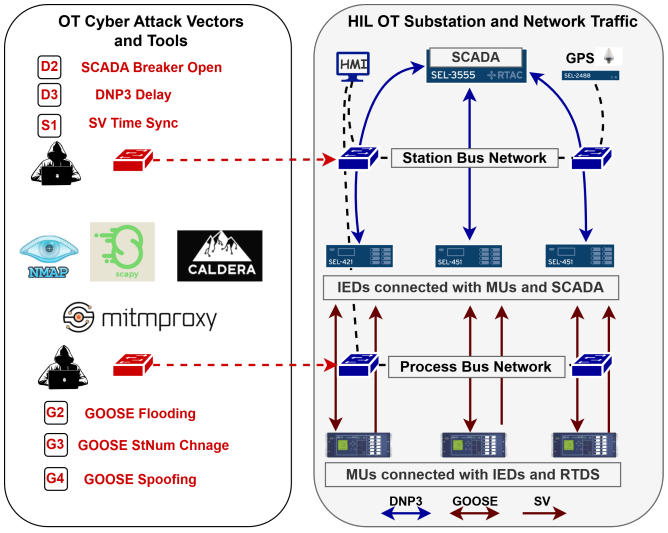


Fig. 1. Hardware in the loop OT substation testbed, communication architecture, and attack vectors and tools used for data generation across the station bus and process bus

we developed a hardware-in-the-loop (HIL) OT substation testbed and a multi-protocol attack dataset spanning deterministic and dynamic attack vectors, and designed a hybrid detection framework combining protocol-aware rule evaluation with graph-based spatiotemporal learning [3]. That framework demonstrated that combining deterministic and learning-based detection substantially improves coverage over either approach alone. However, like most existing OT IDS work, its output remains a single binary decision of anomalous or normal per observation window. It does not indicate which device is compromised, which protocol carries the attack signature, or what mitigation an operator should apply. In safety-critical substation operations, this gap between detection and actionable response limits real-world deployability. Addressing these limitations, this paper makes the following key contributions:

- Proposes an attention-guided root-cause diagnosis mechanism that localizes detected OT anomalies to the responsible device, protocol, and time window.
- Develops a resilience-aware response mapping that translates localized attack signatures into MITRE ATT&CK for ICS-aligned, operator-actionable mitigation.
- Integrates cyber anomaly detection, diagnosis, and re-

sponse into a unified multi-protocol OT cyber-resilience framework validated through HIL substation environments and cyber-resilience metric.

II. Testbed and Operational Flow

The hardware-in-the-loop (HIL) power grid OT substation testbed used in this study is illustrated in Fig. 1. The platform integrates SCADA/HMI, SEL IEDs, merging units, and an RTDS-based power-system simulator. Station- and process-bus networks provide communication among the substation devices. DNP3, IEC 61850 GOOSE, Sampled Values (SV), and PTP support the principal OT communication flows. The architecture builds on our previously developed cyber-resilient IEC 61850 substation environment [4].

The testbed was exercised using multiple protocol-specific OT attack vectors shown in Fig. 1. These include SCADA breaker-open commands, DNP3 delay, SV timing manipulation, GOOSE flooding, *stNum* modification, and spoofing. Attack execution combines MITRE CALDERA with custom protocol-oriented attack implementations and network manipulation tools, enabling controlled and repeatable adversarial scenarios aligned with OT threat behavior [5]. Network traffic was captured during normal and attack operating conditions to generate labeled multi-protocol OT data. The resulting dataset covers seven deterministic and dynamic attack classes and provides the input for the proposed detection and root-cause diagnosis framework.

III. Proposed Methodology

We propose a two-stage theoretical framework that extends spatio-temporal anomaly detection with interpretability and actionability

A. Stage 1: AI-Driven Detection via Graph Neural Networks

OT communication within each time window t is represented as a graph, where nodes represent substation devices and edges represent communication flows. Node representations are updated through edge-conditioned message passing, in which messages between neighboring devices are conditioned on the communication features of the edge that connects them:

$$h_v^{(t)} = \sum_{u \in \mathcal{N}(v)} W(e_{uv}^{(t)}) x_u^{(t)} + W_{root} x_v^{(t)} \tag{1}$$

where $W(\cdot)$ is a learnable transformation conditioned on the edge features and W_{root} is applied to the node itself. A graph-level embedding is then obtained by pooling over all updated node representations:

$$z^{(t)} = \text{concat}(\text{MeanPool}(h_v^{(t)}), \text{MaxPool}(h_v^{(t)})) \tag{2}$$

A recurrent temporal module with attention processes the resulting sequence of graph embeddings across a fixed window and reconstructs an expected normal embedding z_{rec} from the attended context vector. Deviation from this reconstruction defines the anomaly score:

$$S(x) = \|z_{rec} - z_{target}\|_2 \tag{3}$$

which forms the detection signal used to flag anomalous windows. The model is trained exclusively on normal traffic so that it learns a baseline of legitimate multi-protocol behavior.

B. Stage 2: Attention-Guided Root Cause Localization

Where Stage 1 yields only a scalar anomaly score, stage 2 proposes decomposing that score into interpretable components. Node-level contributions are recovered by comparing each node embedding against its learned normal baseline h_v^{norm}:

$$c_v = \|h_v^{(t)} - h_v^{norm}\|_2 \tag{4}$$

which ranks the six substation devices by their contribution to the anomaly. The time window carrying the deviation is localized using the attention weights α_i already computed in Stage 1:

$$t^* = \arg\max_i \alpha_i \tag{5}$$

Where an anomaly is flagged by a violated protocol invariant rather than the learned model, the invariant itself, such as an inconsistent GOOSE state transition, provides a directly interpretable cause with no additional attribution required. The result is a localization signature of device, protocol, and time window attached to every detection.

C. Stage 3: Mitigation driven by Cyber-Resilience metrics

Stage 1 yields anomaly score, stage 2 proposes root cause analytics, stage 3 provides resilience-metric driven mitigation insipired by metric discussed in our past work [3].

IV. Summary

This work presents an interpretable OT intrusion detection framework that extends anomaly detection toward root-cause diagnosis and resilience-aware response. Using the established multi-protocol HIL substation testbed, the framework combines graph-based spatio-temporal detection with attention-guided localization of the affected device, protocol, and time window. The localized attack signature is subsequently associated with an appropriate resilience action, providing a structured path from anomaly detection to operator-actionable response. The proposed framework aims to improve the interpretability and operational usefulness of AI-driven cybersecurity monitoring in power-system OT environments.

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